

Integration Testing

The objective of integration testing is to ensure that the components, while individually functioning correctly continue to do so when they are combined with each other to achieve the end requirement.

System Testing

This test is carried out by interfacing the hardware and software components of the entire system (that have been already unit tested and integration tested), and then testing it holistically.

Acceptance Testing

This test involves making sure that the product developed has met all the requirements and operates as desired by the end user.

Non-Functional testing

Another facet of testing is Non-functional testing, where the non-functional aspects are tested. These aspects are not directly related to the features of the program but might cause it to behave in an abnormal manner.

Non-functional testing involves testing performance, vulnerability to hacks and compatibility on a variety of environments.

Some of the techniques of testing non-functional requirements are:

- Performance testing
- Security testing
- Usability testing
- Compatibility testing

Performance, Load, Stress Testing

There are several variants of performance testing in most testing methodologies, for example: performance testing is quantifying how a system behaves under an increasing load (both numbers of users and data volumes), load testing is verifying that the system can operate at the required replication times when subjected to its expected load, and stress testing is finding the failure point(s) in the system.

Below are some of the most widely used performance testing implements for quantifying web application performance and load stress capacity.

- WebLOAD
- LoadView
- Apache JMeter